AOP (Aspect-Oriented Programming)

FortuneCloud
We Understand Technology

- AOP (Aspect-Oriented Programming) is a programming paradigm in Spring that helps you separate **cross-cutting concerns** (like **logging**, **security**, **transactions**, **etc**.) from your main business logic.

- AOP allows you to add extra behavior to existing code without modifying the code itself.

- AOP makes our application **more flexible**, **less repetitive**, **and easier to manage** as it grows.

Example

1. Create a new java project
2. Add AOP jars

Student.java

```
package Demo;


public class Student {
    public void display() {
        System.out.println("Student details...");
    }
}
```

LoggingAspect.java

```java
package Demo;
import org.aspectj.lang.annotation.After;
import org.aspectj.lang.annotation.Aspect;
import org.aspectj.lang.annotation.Before;

@Aspect
public class LoggingAspect {
    @Before("execution(* Demo.Student.display(..))")
        public void beforeDisplay() {
                System.out.println("Before displaying student details");
    }

@After("execution(* Demo.Student.display(..))")
    public void afterDisplay() {
        System.out.println("After displaying student details");
    }
}
```

Test.java

```
ApplicationContext context = new
ClassPathXmlApplicationContext("applicationContext.xml");

Student s = context.getBean(Student.class);
s.display();
```

Xml File

```xml
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:aop="http://www.springframework.org/schema/aop"
xsi:schemaLocation="
http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd
http://www.springframework.org/schema/aop
http://www.springframework.org/schema/aop/spring-aop.xsd">

<bean id="student" class="Demo.Student"/>

<!-- Aspect -->
<bean id="loggingAspect" class="Demo.LoggingAspect"/>

<!-- Enable AOP -->
<aop:aspectj-autoproxy/>


</beans>
```